

What is SonarQube?

SonarQube is a tool for continuous inspection of code quality. It helps developers maintain high-quality code by identifying bugs, vulnerabilities, and code smells.

In a typical workflow, developers will store their code in a Git repository. They can build the code using Jenkins and Maven, deploying it to Tomcat. Before deployment to a server, it's essential to perform a code inspection to ensure quality and compliance with coding standards. SonarQube facilitates this by analyzing the code and providing detailed reports.

Key Features of SonarQube:

- **Open Source:** SonarQube was introduced by SonarSource as an open-source tool.
- **Integration with Jenkins:** SonarQube can be easily integrated with Jenkins to automate code quality checks.
- **Language Support:** It supports multiple programming languages and provides detailed analysis reports.

Setup Steps:

1. **Launch EC2 Instances:** First, launch two EC2 instances—one for SonarQube and one for Jenkins.
2. **Connect to the Servers:** Access your servers.
 - Jenkins server is accessible via port **8080**.
 - SonarQube server is accessible via port **9000**.
3. **Install Java:** SonarQube requires Java to run.

Add the SonarQube Repository: Use the following command to download the repository:

bash

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```
sudo wget -O /etc/yum.repos.d/sonar.repo
```

```
http://downloads.sourceforge.net/project/sonar-pkg/rpm/sonar.repo
```

4.

Clean YUM Cache:

bash

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```
sudo yum clean all
```

5.

Install SonarQube:

bash

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```
sudo yum install sonar
```

6.

Start and Enable SonarQube:

bash

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```
sudo systemctl start sonar
```

```
sudo systemctl enable sonar
```

7.

Check SonarQube Status:

bash

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```
sudo systemctl status sonar
```

8.

9. **Access SonarQube:** Open your web browser and navigate to your server's IP followed by port **9000**.

Login Credentials:

You can log in using the default credentials:

- **Username:** admin
- **Password:** admin

Create a Token:

Generate a token for your project (e.g., **batch13 : 7eac4d7307a4b2528e88f5544d7117b8dfb34811**).

Configure Your Analysis:

1. Select your programming language.
2. Choose your build tool (e.g., Maven).

SonarQube will provide an automatic command to scan with your build tool. Example:

bash

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```
mvn sonar:sonar \
  -Dsonar.host.url=http://<your-server-ip>:9000 \
```

```
-Dsonar.login=7eac4d7307a4b2528e88f5544d7117b8dfb34811
```

3.

Code Analysis:

You can clone your repository and use the above command to check the quality of your code. SonarQube will display your code review results, categorizing issues as blocker, minor, and critical.

Code Review Process:

You can share the link generated after executing the command with your developers. Based on the alerts, you can provide feedback on the code quality, focusing on blocker and critical issues, which should be addressed promptly.

This manual process outlines the setup of SonarQube for code quality checks.